

# CentOS DHCP and DNS Server Administration Lab

## Project Objectives

### 1. Primary Objective

The primary objective of this project is to deploy and configure a CentOS Linux server to provide centralized DHCP and DNS services within a virtualized network environment.

### 2. Specific Objectives

The project aims to achieve the following objectives:

1. Install and configure CentOS Linux as a network services server.
2. Configure a static IP address for the CentOS server.
3. Configure the VMware Host-Only network for the project environment.
4. Install and configure a DHCP server using the `dhcpd` service.
5. Create a DHCP address pool for automatically assigning IP addresses to client systems.
6. Configure DHCP to provide clients with network information including IP address, subnet mask, gateway, DNS server, and DNS domain.
7. Install and configure the BIND DNS server.
8. Create a forward DNS zone for the `lab.local` domain.
9. Create a reverse DNS zone for the `192.168.10.0/24` network.
10. Configure DNS A records for the server and client systems.
11. Configure DNS PTR records for reverse DNS resolution.
12. Integrate DHCP and DNS services.
13. Configure CentOS client systems to obtain network settings automatically through DHCP.
14. Configure clients to use the CentOS server at `192.168.10.2` as their DNS server.
15. Test forward DNS resolution from hostname to IP address.
16. Test reverse DNS resolution from IP address to hostname.
17. Verify communication between the server and client systems.
18. Verify communication between the client systems.
19. Monitor and verify DHCP and DNS services.
20. Troubleshoot network, DHCP, and DNS-related problems.
21. Document configuration procedures, testing results, and troubleshooting solutions.
22. Develop practical Linux server administration and network services skills.

### 3. Learning Outcomes

After completing this project, the following skills will be demonstrated:

- CentOS Linux administration

- VMware virtual networking
- Static IP configuration
- DHCP server administration
- DNS server administration
- BIND configuration
- Forward DNS configuration
- Reverse DNS configuration
- DHCP-DNS integration
- Client network configuration
- Network troubleshooting
- Service management using systemctl
- Firewall configuration
- Log analysis
- Technical documentation

## 4. Expected Result

At the completion of the project:

- The CentOS server will operate as a DHCP server.
- The CentOS server will operate as a DNS server.
- Clients will automatically receive IP addresses through DHCP.
- Clients will automatically receive DNS server information.
- Internal hostnames will resolve successfully.
- Reverse DNS lookups will work successfully.
- Server and client connectivity will be verified.
- All configurations and troubleshooting activities will be documented.